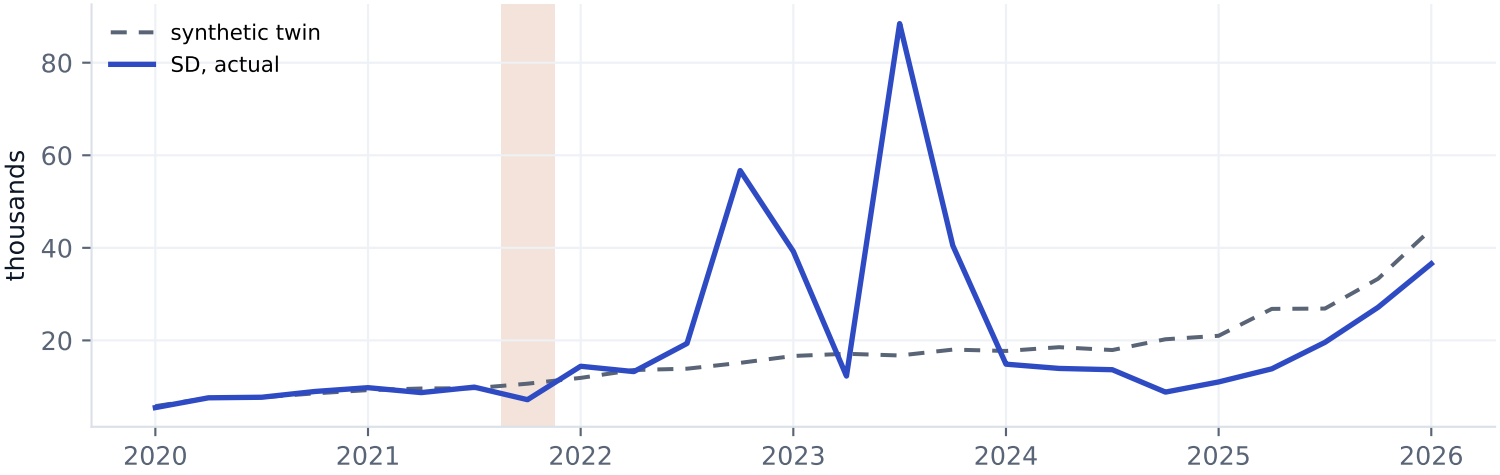
Sudan, 2021

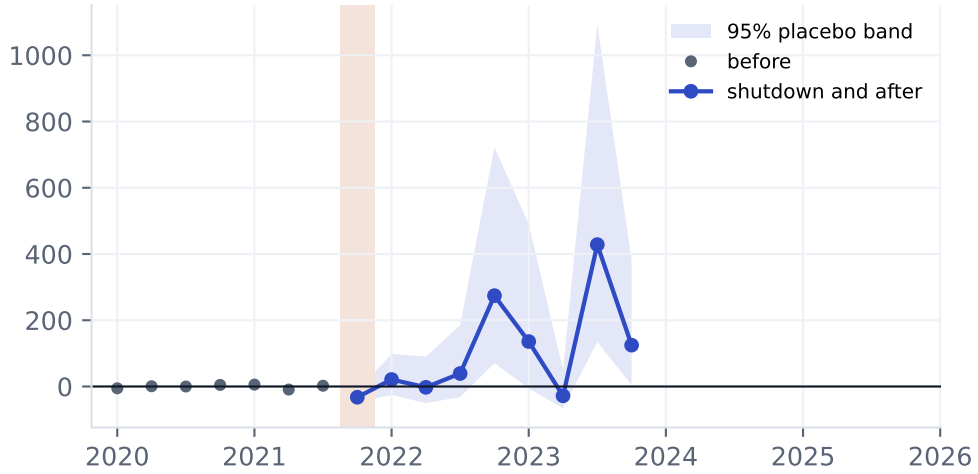
Change in git pushes attributed to SD in 2021Q4 vs a synthetic twin (106 donor economies, 7 pre quarters)

Method	vs twin	95% interval	p normal	p perm
Synthetic DiD	-32.5%	-47% to -13%	0.002	0.019
Synthetic control	-38.4%	-53% to -19%	<0.001	0.019
Augmented synthetic control	-41.9%	-54% to -26%	<0.001	0.009

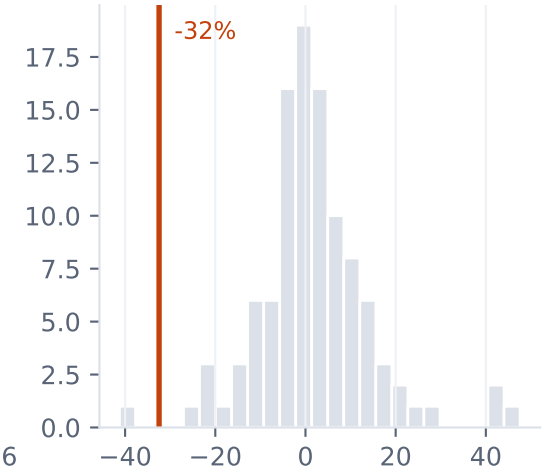
Git pushes per quarter, attributed to SD



Gap to the twin (%)



Same design run on every donor (%)



- Conformal test. p = 0.125. With 8 quarters the smallest possible p is 0.125, so this test can't give an interval here.
- Backdating. Fitting without 2021Q2 and 2021Q3, the twin misses those quarters by -25%, -11%. Outside the placebo band, so the pre trend is shaky.
- Breakdown. The twin's path would have to be off by 1.5 times the worst miss in the pre period fit (a weaker benchmark, there is too little history for fake events) before the result goes away.
- Leave one donor out. Dropping each of the 30 donors with weight over 1% keeps the estimate between -33.6% and -31.0%.
- Holm adjusted p across the nine registry events is 0.015. Does not pass every check in the analysis plan, so I don't call it a clear drop.
- What happened. Fixed and mobile internet cut on 25 Oct 2021 after the coup, services began returning 18 Nov 2021. Source https://pulse.internetsociety.org/en/shutdowns/internet-disruption-in-sudan/

GitHub activity placed in the country by IP address, not national output. VPN users get placed where the VPN exits.